Azzar Budiyanto

Freelance Software Developer · IoT · Full-Stack · AI Tooling · Automation · Consulting · Teaching

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PROFILE

Engineering resilient digital ecosystems — from embedded firmware and control systems to AI-powered web platforms and the automation tooling that ties them together. Freelance software developer with 8+ years delivering production systems end-to-end: IoT firmware, cloud pipelines, full-stack web applications, AI agent tooling, browser automation, and consulting services across SEO, web strategy, and technical training. Author of 127+ public repositories and 13 live commercial deployments. Multilingual, remote-first, and pragmatic about scope, deadlines, and documentation.

CORE COMPETENCIES

PROGRAMMING	Python, TypeScript, JavaScript, C/C++, Rust, Go, Bash, SQL, MATLAB
WEB STACK	React, Next.js, Vue.js, Node.js, Express, Django, Flask, Tailwind, WordPress, REST/GraphQL
EMBEDDED & IOT	ESP32, STM32, Arduino, Raspberry Pi, LoRa, MQTT, Wokwi, Tinkercad, Sensor/Actuator Integration
CONTROL SYSTEMS	PID, Fuzzy Logic, LQR, Ziegler-Nichols, Cohen-Coon, Genetic Algorithms
DEVOPS & CLOUD	Docker, Kubernetes, AWS, GCP, Azure, Firebase, Terraform, CI/CD, NGINX
AI / AUTOMATION	MCP Servers, Agent Skills, Playwright, Puppeteer, TensorFlow, PyTorch, Scikit-learn, OpenCV, NLP, Generative AI
DATA & DATABASES	PostgreSQL, MySQL, SQLite, MongoDB, Redis, Elasticsearch
CONSULTING & SEO	Technical SEO, Core Web Vitals, Structured Data, On/Off/Technical Page, Web Strategy, AI-Optimized Content
UI/UX & DESIGN	Figma, Tailwind, KiCad, PCB Design, AutoCAD, Photography

EXPERIENCE

Full Stack Engineer — *PT. Jejaring Media Arus Utama* (Aug 2026 – Present)

- Built and maintained a multi-service media platform using Go, Gin, Bun ORM, PostgreSQL, React, Astro, Tailwind CSS, Flutter, and Cloudflare infrastructure across web, admin, portal, mobile, and API services.
- Designed and implemented database connectivity hardening for Cloud Run API connected to Neon PostgreSQL (pool tuning, timeout handling, transient error classification, safe 503s, staging min-instances).
- Diagnosed intermittent production database timeouts by correlating Cloud Run runtime behavior, Neon PostgreSQL connectivity, IPv4/IPv6 paths, connection pooling, and cold-start scenarios.
- Established a structured staging-to-production delivery workflow using GitHub PRs, Actions, Cloud Run, Cloudflare Workers, and Pages with repository parity checks and rollback-aware release practices.
- Improved system reliability through encrypted vault credential management, staging-only intensive testing, no-blind-retry policies for database writes, and SSR resilience.

Lead Developer — *PT. Turga Berkah Kubra* (Jan 2026 – Nov 2026)

- Led full-stack web development projects, architecting scalable solutions from concept to deployment.
- Integrated AI/ML models into production applications, enhancing automation and decision-making capabilities.
- Managed DevOps infrastructure, implementing CI/CD pipelines and containerized deployments.
- Oversaw development team, conducting code reviews and ensuring code quality standards.
- Collaborated with stakeholders to define technical requirements and project roadmaps.

Freelance Software Developer — *Self-employed, Remote Worldwide* (Oct 2018 – Present)

- Deliver end-to-end IoT and web projects: firmware, cloud pipeline, dashboard, deployment — 13 live commercial sites across travel, news, energy, e-commerce, and SaaS.
- Design control algorithms (PID, Fuzzy, LQR) for industrial and research prototypes.
- Author MCP servers and agent skill libraries extending Claude and OpenAI-compatible models.
- Build browser automation pipelines (Playwright, Puppeteer) for data extraction and visual regression.
- Architect REST and GraphQL APIs in Node.js, Django, and Flask; deploy on AWS, GCP, and Azure.
- Build responsive front-ends in React, Next.js, and Vue.js with Tailwind; integrate WebSocket telemetry and AI-powered content pipelines.
- Deliver web, SEO, and AIO consulting: technical SEO audits, Core Web Vitals optimisation, structured data, and AI-assisted content strategies.

Microcontroller Trainer — *Indobot Academy, Yogyakarta* (Jan 2022 – Jan 2023)

- Designed curriculum for Arduino, ESP32, and STM32 tracks; taught 4 cohorts of 20+ participants.
- Delivered IoT modules covering MQTT, cloud telemetry, and dashboard integration.
- Provided 1:1 mentoring; produced learning-progress reports and certification documents.

Photographer — *Lembaran Kasih, Tuban* (Sep 2020 – Sep 2021)

- Shot high-quality stills for print and digital campaigns under tight deadlines.
- Edited, toned, captioned, and uploaded deliverables for publication pipelines.

Student Ambassador — *cicil.co.id, Yogyakarta* (Jan 2020 – Apr 2021)

- Streamlined student registration, technical onboarding, and documentation flow.
- Resolved administrative and account issues directly with end-users.

SELECTED PROJECTS

LIVE COMMERCIAL — 13 SITES

Client Web Portfolio

Production deployments across travel (Syaz Travel), news (Sindikasi Media, Berandawirson), energy (New Energi), livestock (Top Global Farming), HR (HYStory), SaaS tools, and more.

AI TOOLING — CLOUDFLARE WORKERS

Live Chat Widget + AI Blog

TypeScript chat widget served via Cloudflare Workers; AI-powered automated blog running MEMA AI content pipeline with scheduled publish hooks.

OPEN SOURCE — 170 STARS

use-waydroid-on-x11

Most-starred repository — guides for running Android containers on X11 Linux systems; widely adopted in the Linux community.

IOT / INDUSTRIAL

Multi-Sensor Telemetry Stack

ESP32 firmware with SHT20, thermistor, and MQ-137 sensors streaming to MQTT; Node.js + React dashboard for live and historical data.

CONTROL SYSTEMS

Fuzzy + PID Temperature Loop

Auto-tuned PID (Ziegler-Nichols, Cohen-Coon) with fuzzy supervisor for non-linear regimes. Simulated in MATLAB, deployed on STM32.

CERTIFICATIONS

IoT Practitioner — BNSP (2024)

Belajar Fundamental Deep Learning — Dicoding (2023)

Membangun Sistem Machine Learning — Dicoding (2023)

Belajar Machine Learning untuk Pemula — Dicoding (2023)

Belajar Dasar AI — Dicoding (2023)

Memulai Pemrograman dengan Python — Dicoding (2023)

Problem Solving (Intermediate) — HackerRank (2023)

Problem Solving (Basic) — HackerRank (2023)

Python (Basic) — HackerRank (2023)

C++ — HackerRank (2023)

SQL (Intermediate) — HackerRank (2023)

SQL (Basic) — HackerRank (2023)

CSS — HackerRank (2023)

REST API (Intermediate) — HackerRank (2023)

Build Your Beautiful Apps Using SwiftUI — Dicoding (2020)

Scrum Introduction — Dicoding (2020)

Lean-Agile Mindset — Dicoding (2021)

GitHub Achievements — Starstruck x2, Pull Shark x2 (2023–2025)